

RESUME . PDF

filippofonseca.com



I build *humanist physical intelligence* for the human body.

Striving to enable people to live healthier, more fulfilling lives through life-saving, enriching, physically-intelligent systems.

- CARDIOVASCULAR & ORTHOPEDIC MECHATRONICS
- SOFT ROBOTICS
- ACCESSIBILITY TECH
- EMBEDDED AI/ML
- CLOUD/FULL-STACK/IOS DEV

YALE '28 | MECHANICAL ENGINEERING (ABET) & EECS (DOUBLE MAJOR)

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I'm a Yale junior double-majoring in **ABET Mechanical Engineering** and the **EECS (Electrical Engineering + Computer Science)** joint major. I'm an entrepreneur and biomechatronics & cardiovascular engineer, focused on **building humanist healthcare and physical intelligence through robotics, materials, and ML**. Particularly interested in cardiovascular engineering, but also work extensively in orthopedics and prosthetics. I originate from Italy 🇮🇹, Spain 🇪🇸, & Costa Rica 🇸🇨. Over the years, I've founded several software- and hardware-based ventures and conducted extensive research at leading laboratories worldwide. My broader interests span marathon running, linguistics (polyglot), Stoic philosophy, writing, and theoretical physics.

EDUCATION



B.S. in ABET Mechanical Engineering & EECS (Electrical Engineering + Computer Science)
Yale University | New Haven, CT, USA

2024 - 2028

WORK EXPERIENCE



Biomedical Engineering Intern: Biomechanics and Biomaterials

Harvard Medical School / Massachusetts General Hospital

May 2026 - Present

- Harris Orthopaedics Laboratory—developing next-generation implant materials, devices, and simulations for orthopedic surgery
- Reducing post-operative infection risk through mechanical devices leveraging pharmacokinetic/pharmacodynamic (PK/PD) optimization
- Simulating intra-articular (in joint) conditions for local antibiotic delivery



Cardiovascular Biomechanics and Tissue Engineering - Undergraduate Researcher

Yale Integrative Cardiac Biomechanics Laboratory

Feb 2025 - Present

- Engineering Dynamic Electromechanical Bioreactor (DEB) replicating exercise stress on human-engineered heart tissues (hEHTs)
- Studying Arrhythmogenic Cardiomyopathy progression in endurance athletes
- Developing programmable electromechanical loading system with real-time force feedback + electrophysiological readouts



President

Yale Undergraduate Robotics

Aug 2024 - Present

- First sophomore ever selected to lead a venture at Yale's largest engineering society
- Directing 30+ engineers across mechanical, robotics, science, and E&C subteams for URC 2026 (Utah)
- Tech: ROS2, sensor fusion, LiDAR, ML/TensorFlow, SLAM, pose estimation, computer vision
- Leading design reviews, test campaigns (mobility, manipulator, bio payloads), system-level risk mitigation



Summer Research Fellow, Reactor Engineering

UCLouvain - Université catholique de Louvain

Jun - Aug 2025

- Simulated CO₂ methanation reactors under concentrated streams; evaluated methane yield & thermal runaway thresholds
- Built heterogeneous multitubed + multitubular fixed-bed reactor models
- Applied CFD, MATLAB, Python for dynamic simulations; improved parameter estimation via model reduction
- Delivered technical reports on flue gas cleaning + energy efficiency under Prof. Juray De Wilde



Robotics Engineer

The Laboratory at Yale (Kramer-Bottiglio Lab)

Apr 2025 - Present

- Bio-inspired burrowing robots for planetary exploration and environmental sensing
- Designing soft robotic actuators, locomotion tests, control systems for granular media navigation

**Mechanical Design Engineer, Precision Microtome Device**

Spring 2026

Johnson & Johnson Vision Care

- Selected to design and build a precision microtome-style device capable of producing ~10 µm reproducible sections of soft, compliant polymers without embedding or cryogenic stiffening
- Developing custom clamping, micron-scale feed mechanisms, and blade kinematics to minimize compression, shear, and deformation during sectioning
- Engineering a non-contaminating mechanical sectioning workflow to enable accurate refractive-index depth profiling across the first 100–1000 µm of material
- Validating section thickness accuracy, flatness, and repeatability using optical microscopy and mechanical calibration methods

**Lead, UCT Algorithmic Benchmarking**

Aug 2024 - Sept 2025

US Space Command / Yale Space Policy & Research Collaborative

- Spearheaded Yale's collaboration with U.S. Department of Defense (Space Command) on space situational awareness
- Led cross-disciplinary research on satellite UCT algorithms coordinating government, academia, industry
- Results accepted at AMOS 2025 (Maui, Hawaii) — premier global space domain awareness conference

**Mobile App Engineer**

Sept 2025 - Present

Yale Department of Political Science / Government of Greece

- Mobile app for Greek government aiding migrants/asylum seekers: legal rights, hotlines, job matching
- Cross-platform stack (React Native, TypeScript) serving thousands of end-users across Europe
- Bridging Yale research with real-world humanitarian impact

**Platform Engineer**

Sept 2024 - Aug 2025

CourseTable

- One of ~10 Yale undergrads selected; most-used website at Yale (~95% student body, millions of semester requests)
- Engineered professor analytics insight page, optimized calendar view, high-reliability features

**Frontend Systems & Architecture Engineer**

May - Aug 2024

VoqoAI Australia

- Led frontend systems design for B2B AI calling platform backed by Australia's largest investors
- Built core features from scratch (agent management, call logs) from Figma to full-stack
- Distributed team collaboration across 16-hour time difference; React, TypeScript, Next.js, MongoDB, Docker

● VENTURES & STARTUPS

**Founder**

Apr 2025 - Present

DegreeIntelligence

- Yale's leading degree-tracking platform — used by 1 in 7 undergraduates (~750+ users)
- Built full-stack entirely solo: Next.js, Firebase, OpenAI APIs
- AI advising + collaborative planning features

**Co-Founder, Lead Engineer (Acquired)**

Apr - Nov 2023

HSC-GPT → Acquired by Art of Smart Education

- Australia's first localized AI education platform for national curriculum (HSC)
- Adopted by thousands of students, outpacing edtech incumbents
- Led engineering of AI-powered adaptive learning system; acquired 2023

**Founder & Lead Engineer**

Jul 2022 - Jan 2024

Diush

- Open-source Gen-Z social commerce app; 5,000+ users on waitlist
- Raised VC funding from 1517 Fund (Medici Grant)
- Won 1st place in AASCA international entrepreneurship competition (Central America)
- Drove design, engineering, and growth as sole maintainer

**CEO & Co-Founder**

Jan - Nov 2021

Zyndicate

- All-in-one productivity ecosystem ("LifeOS") for entrepreneurs and teams
- Scaled to 31,500+ users (10,500 weekly active) by age 16
- Recruited and led 15+ member international team across 8 countries
- VC conversations with Andreessen Horowitz; explored acquisition paths

**Co-Founder**

Jan - Nov 2021

UpDrop Drone Systems (World Robot Olympiad)

- At age 15, co-engineered autonomous drone delivery system for WRO Open Category
- 1st place Costa Rica, 2nd worldwide out of 14,000+ teams
- Featured in international media (La Nación, Teletica, El Mundo)
- Commended by President of Costa Rica with personal mention and public statement

Creator

YouTube Channel: "Filippo Studies"

→ 200,000+ total views, 1,500+ subscribers

→ Content: learning systems, productivity frameworks, polyglot strategies (7+ languages)

→ Featured: "19-year-old polyglot speaks in 7 languages", "Reading ALL THE ESSAYS that got me into Yale"

Mar 2023 - Present

PAPERS & PUBLICATIONS



Uncorrelated Track (UCT) Processing: Establishing Efficient Algorithmic Treatment and Benchmarking Best Practices for Space Domain Awareness (SDA)

AMOS Conference 2025 · Maui, Hawaii

F. Fonseca, O. Pinhasi, Z. Zitzewitz · Paper · September 2025

Published



Standardizing Experimental Setups for Robotic Locomotion in Granular Media

IEEE RoboSoft 2026 · Japan

C.L. Le, F. Fonseca*, N. Ramos*, E.I. Figueroa, C. Creager, R. Kramer-Bottiglio · Paper · April 2026

Published

AWARDS

- **Y Combinator Startup School 2026**, selected as one of a few builders out of 30,000+ applicants worldwide (attending in San Francisco, July 2026)
- **Google x Yale SOM Hackathon 2026** — 2nd Place Overall with **iSpy**, a wearable AI vision system for the blind and visually-impaired
- **YHack 2026** — 5th Place Hardware at Yale's premier national hackathon with **PosturePoke**, a posture wearable that fights slouching with escalating punishments
- **World Robot Olympiad 2021** — 2nd Place Worldwide (14,000+ teams); 1st Place Costa Rica
- **1517 Fund Medici Grant** — VC grant recipient
- **HSC-GPT Acquisition** — Platform acquired by Art of Smart Education (2023)
- **AASCA Entrepreneurship** — 1st Place, international competition (Central America)

TECHNICAL SKILLS

Engineering: Mechatronics · Robotics · Soft Robotics · Reactor Modeling · CFD · PCB Design · Power Systems

Software: TypeScript · Python · React/Next.js · Firebase · MATLAB · C/C++ · Java · Flutter · Rust · Git · Docker · ROS2

AI/ML: TensorFlow · PyTorch · Computational Modeling · Data Analysis · Regression · Predictive Modeling · OpenAI APIs

Hardware: SolidWorks · Fusion 360 · Arduino · Raspberry Pi · Embedded Systems · PCB Design · 3D Printing

Leadership: Product Management · Technical Writing · Team Leadership (15+ reports) · Cross-Disciplinary Collaboration

LANGUAGES

 **English** · Native

 **Italian** · Native

 **Spanish** · Native

 **French** · Fluent

 **Bulgarian** · Conversational

 **Portuguese** · Conversational

 **Norwegian** · Learning